

A Diagnostic No-Go for Train-Free Localization Probes in Attribute-Object Binding

Anonymous submission

Abstract

Vision-language systems can choose the right caption in paired image-text tests without revealing whether they bind each attribute to the intended object. We present a diagnostic no-go study of a train-free way to test that bridge: frozen scorers and available object evidence are used to compare full-image, no-region local, detector-region, and oracle-region lanes across NLVR2, SugarCrepe, SUGARCREPE++, and VAW, with blind and bag controls plus a Qwen/Qwen2.5-VL-3B-Instruct reference lane. Across 7,420 completed scored trials, correct VAW regions produce the clearest localization-sensitive signal, raising object-local OpenCLIP accuracy from 0.543 without regions to 0.823 with oracle regions. The practical non-oracle bridge remains weak under the current benchmark schema: detector evidence is VAW-only, reaches 0.595 VAW accuracy, matches 321/880 region rows, leaves 559 fallback or missing rows, and yields recall@0.5 of 0.158, while the selected non-VAW families do not provide source regions. Together with no-region local scoring tying global OpenCLIP on NLVR2, SugarCrepe, and SUGARCREPE++ and a Qwen result limited to a 320-row reversed-slice audit, these results support a calibrated negative conclusion: localization can matter, but this train-free route is not yet a practical localization method without new region evidence or a full position-balanced VLM audit.

Introduction

Compositional vision-language evaluation asks whether a model connects the right visual evidence to the right words rather than merely recognizing that all mentioned objects and attributes are present (Thrush et al. 2022). This distinction has become central in diagnostic benchmarks for visual reasoning and image-text matching: CLEVR isolates synthetic compositional reasoning (Johnson et al. 2017), Winoground holds word sets fixed while changing word order (Thrush et al. 2022), and SugarCrepe shows that benchmark artifacts can make hard negatives hackable (Hsieh et al. 2023). Attribute-object binding is a particularly sharp case. A model can know that a scene contains a red object and a square object while still choosing the caption that binds the adjective to the wrong noun (Yuksekgonul et al. 2023). Large image-caption resources such as COCO (Lin et al. 2014), Flickr30k (Young et al. 2014), and VQA (Antol et al. 2015) made broad multimodal training and evaluation possible, but broad supervision does not by itself certify bind-

ing. Region-aware pretraining lines such as OSCAR (Li et al. 2020), VinVL (Zhang et al. 2021), and UNITER (Chen et al. 2020) further show why object evidence and language evidence should be audited together rather than assumed to align.

The problem is not only whether large vision-language models can answer a particular benchmark item. It is also whether an evaluation can separate binding evidence from shortcuts. Prior work has shown that contrastive vision-language models can behave like bags of words (Yuksekgonul et al. 2023; Yuksekgonul, Bianchi, and Zou 2022), that text encoders can bottleneck compositionality (Kamath, Hessel, and Chang 2023), and that benchmark negatives can overstate progress when blind or lexical controls succeed (Hsieh et al. 2023; Udandara et al. 2025). Those findings motivate a measurement question: when a frozen scorer fails or succeeds on binding alternatives, is the outcome driven by global image-text alignment, object-local evidence, an oracle localization upper bound, or a response-order artifact, a concern also reflected in checklist-style evaluations (Zhao et al. 2022)?

We answer that question with a train-free localization-probe design. Each item keeps a correct caption and a minimally changed binding alternative, while the scoring lane changes what visual evidence the frozen model can use. Global OpenCLIP scores the full image; no-region object-local OpenCLIP tests whether a local readout helps without explicit regions; a VAW-only detected-region lane uses a COCO-pretrained Faster R-CNN detector before OpenCLIP scoring; a VAW-only oracle lane supplies gold object boxes; blind and bag controls test shortcut explanations; and a Qwen/VLM lane tests a stronger instruction-tuned answerer. Figure 1 summarizes how the lanes feed paired tests and claim interpretation.

The measurement gives a compact empirical map of what survives these controls. On VAW, oracle regions improve object-local OpenCLIP by 0.2795 accuracy over the same no-region readout, showing that correct localization can change the binding decision for a frozen scorer. The automatic detector lane recovers a smaller part of that gap: it improves VAW accuracy by 0.0523 over no-region object-local scoring, but its repaired coverage audit still matches only 321/880 region rows and leaves most rows as fallback or missing. The negative result is also informative:

Train-free binding evaluation pipeline

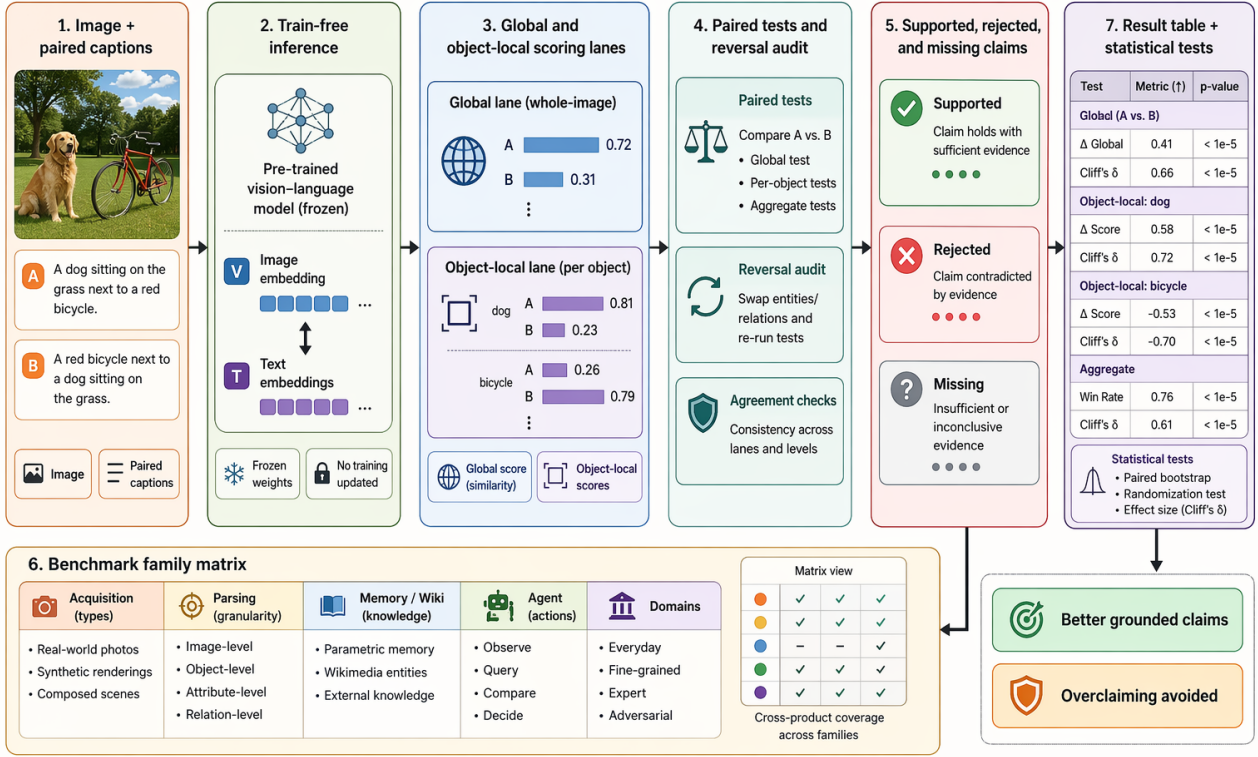


Figure 1: Train-free binding evaluation pipeline. Paired image-caption alternatives first pass through train-free inference and scoring lanes; only paired tests and the reversed-order audit feed the claim summary. The benchmark matrix organizes reporting slices rather than directly creating evidence.

no-region object-local scoring does not beat global OpenCLIP on NLVR2, SugarCrepe, or SUGARCREPE++, so locality without reliable regions is not a general fix. Finally, the Qwen/VLM reference remains high under a 320-row reversed-order audit, but because the full configured-order run places every gold answer first, the paper treats that lane as slice-audited rather than fully position-balanced.

The paper’s central tension is that several plausible explanations can produce the same headline behavior. A full-image contrastive scorer can choose the right caption because the image and text globally align, not because it assigns the attribute to the intended object. A local readout can help because it isolates the object, or it can fail because the required region evidence is not actually available without regions. A large generative VLM can choose the right option because it understands the binding, because one candidate is phrased more naturally, or because the answer position is favorable. The evaluation is built to keep these explanations separated. That is why the paper reports controls, paired tests, oracle regions, and a candidate-order audit in the same evaluation.

This separation determines the scientific claim. The contribution is a diagnostic no-go account of when localization evidence changes a train-free binding decision and why the

current practical route is insufficient. It names each scoring lane and its evidential role, reports family-level paired comparisons before interpretation, and keeps the positive VAW oracle result, the coverage-limited detector result, the non-VAW schema no-go, the negative broad-object-local result, and the Qwen candidate-order limitation in the same evidence ledger.

Related Work

Compositional vision-language benchmarks. Diagnostic datasets have long stressed that multimodal systems need more than object presence. CLEVR provides a synthetic benchmark for compositional visual reasoning (Johnson et al. 2017); NLVR2 grounds natural-language reasoning in paired photographs (Suhr et al. 2019); Visual Genome and VAW provide dense object and attribute annotations that make object-level analysis possible (Krishna et al. 2017; Pham et al. 2021). Winoground directly probes visio-linguistic compositionality with paired images and captions containing identical word sets (Thrush et al. 2022). SugarCrepe and SUGARCREPE++ focus on fixing or extending compositional image-text alternatives (Hsieh et al. 2023; Dumpala et al. 2024). This paper borrows the paired-decision spirit of those benchmarks but uses the local re-

sult matrix to ask which scoring lane supports each claim (Thrush et al. 2022).

Shortcut and bias controls. A recurring lesson is that benchmark accuracy can be inflated by artifacts. FOIL and later stress tests showed that small image-language mismatches can expose failures (Shekhar et al. 2017; Zhao et al. 2022). SugarCrepe demonstrated that blind models can exploit negative-caption artifacts (Hsieh et al. 2023); Good CREPE revisits related biases in compositional benchmarks (Udandarao et al. 2025). ARO and NegCLIP connect such failures to bag-of-words behavior and contrastive shortcuts (Yuksekgonul et al. 2023; Yuksekgonul, Bianchi, and Zou 2022). The blind text-only and global-bag lanes in this study therefore serve as measurement controls, not as competitive methods, following the same caution about artifact-sensitive comparisons (Hsieh et al. 2023).

Vision-language model families. The evaluated frozen scorers sit in a larger lineage of image-text representation learning. CLIP popularized large-scale contrastive supervision (Radford et al. 2021), while OpenCLIP and LAION made reproducible open contrastive scaling more accessible (Cherti et al. 2023; Schuhmann et al. 2022). Earlier cross-modal transformers such as ViLBERT, LXMERT, VisualBERT, UNITER, OSCAR, VinVL, and ViLT explored fusion and region representations (Lu et al. 2019; Tan and Bansal 2019; Li et al. 2019; Chen et al. 2020; Li et al. 2020; Zhang et al. 2021; Kim, Son, and Kim 2021). BLIP, BLIP-2, InstructBLIP, LLaVA, and Qwen-VL/Qwen2-VL show the move toward instruction-following and generative VLMs (Li et al. 2022, 2023; Dai et al. 2023; Liu et al. 2024; Bai et al. 2023; Wang et al. 2024). This paper evaluates only the evaluated model/backend instances measured in this study and avoids claims about the full family of VLMs.

Inference-time and localization interventions. Recent work argues that inference structure can matter for compositionality even when weights are frozen (Miranda et al. 2025, 2026, 2024). Training-time or data-centric methods such as TripletCLIP and iterated learning target stronger compositional representations (Patel et al. 2024; Zheng et al. 2024), while hallucination and object-alignment work studies whether visual evidence is used faithfully (Chen, Liu, and Qiu 2026; Tan et al. 2026; Wang et al. 2026). Our measured contribution is narrower: we test whether object-local evidence helps under frozen scoring, and we separate an oracle localization upper bound from coverage-limited detected-region recovery (Chen, Liu, and Qiu 2026).

The closest methodological precedents are inference-time restructuring papers that decompose image-text matching without retraining (Miranda et al. 2025, 2026, 2024). The closest evaluation precedents are benchmark papers that treat blind baselines, hard negative construction, and slice analysis as first-class evidence (Hsieh et al. 2023; Thrush et al. 2022; Zhao et al. 2022). This study sits at the intersection. It does not introduce a new dataset and does not train a new model; instead, it audits how different frozen scoring lanes behave on a binding-focused benchmark matrix. That makes calibration more important than novelty language. The result is useful precisely because it separates a supported localization-sensitive effect from two tempting

overstatements: broad object-local improvement and unconstrained practical detected-region recovery.

Several related benchmark papers make the same caution necessary. Winoground shows that holding the word set fixed can reveal failures that ordinary caption-retrieval metrics hide (Thrush et al. 2022). SugarCrepe shows that negative construction itself can create exploitable shortcuts (Hsieh et al. 2023). VL-CheckList and ARO organize errors by objects, attributes, and relations rather than by aggregate retrieval alone (Zhao et al. 2022; Yuksekgonul et al. 2023). These papers motivate the study’s interpretation ledger: a single accuracy table is not enough unless it is paired with controls that say what kind of evidence produced the number.

The literature also warns against treating model scale as a complete answer. Instruction-tuned VLMs such as BLIP-2, InstructBLIP, LLaVA, and Qwen-VL variants extend what image-conditioned language models can do (Li et al. 2023; Dai et al. 2023; Liu et al. 2024; Bai et al. 2023; Wang et al. 2024), but benchmark-specific audits remain necessary. A high VLM score in a fixed candidate order is useful but not conclusive. The reversed-order slice in this study is therefore not a cosmetic appendix check; it is part of the evidential contract for any statement about the Qwen/VLM lane.

Method

The evaluation converts each benchmark item into a pairwise decision. A method receives an image and two candidate captions or answers: one preserves the benchmark’s target binding and one changes the binding while keeping much of the lexical content comparable. The metric is pairwise accuracy, averaged by condition and benchmark family. No condition updates model weights; all reported lanes are inference-only.

Protocol specification. The evaluated system is a fixed paired-ranking harness rather than a learned model: each lane receives the same two candidates, uses its configured scorer or answer generator once per item, selects one candidate, and reports family-level accuracy plus paired deltas only when task identifiers are shared. The source families are NLVR2 replacement/robustness items ($n = 354$), SugarCrepe attribute-object swaps ($n = 257$), SUGARCREPE++ attribute-object swaps ($n = 257$), and VAW attribute-object localization items ($n = 440$). The lane set consists of blind text-only and bag controls, global OpenCLIP, object-local OpenCLIP without regions, VAW detector-region OpenCLIP, VAW oracle-region OpenCLIP, and a Qwen/Qwen2.5-VL-3B-Instruct reference lane. Detector and oracle region inputs are available only for VAW in this evidence package; detector abstentions are counted as full-image fallbacks, not dropped items.

Evaluated backends and budget. The OpenCLIP lanes use a ViT-B/32-style contrastive scorer with the LAION2B S34B B79K checkpoint for global image-text scoring, no-region object-local scoring, VAW detected-region scoring, and VAW oracle-region scoring. The blind text-only lane uses the same text-side scoring family as an image-blind control, and the global-bag lane is a deterministic unordered caption-overlap control. The VLM reference lane uses

Qwen/Qwen2.5-VL-3B-Instruct to choose between the two candidates with a temperature-0, eight-token answer budget; the completed configured-order run records 1,308 VLM calls, one per scored item. The completed evaluated lanes are train-free and inference-only; no model weights are updated.

Global scoring lane. The global OpenCLIP lane scores the full image against each candidate text and chooses the higher-scoring candidate. It is the natural frozen dual-encoder baseline: it tests whether whole-image alignment is enough to resolve the binding alternative. The global-bag control removes ordering and binding information in a deterministic bag-style comparison; when both alternatives reduce to the same bag, it exposes tie behavior rather than semantic understanding.

Object-local lane. The no-region object-local OpenCLIP lane uses the same frozen model family but changes the readout to emphasize region-level evidence when explicit regions are unavailable. This condition is important because it tests a practical-sounding decomposition without assuming a detector. The local evidence does not support a broad improvement claim for this lane.

Detected-region lane. The detected-region VAW lane uses a COCO-pretrained Faster R-CNN ResNet-50-FPN detector to select target-object crops when a target object maps to a detector category and falls back to the full image when the detector abstains. It reaches 0.595 VAW accuracy, but the coverage facts are part of the result: the repaired localization-coverage audit expands detector-matched rows from the previous 208/880 baseline to 321/880, leaves 559 fallback or missing rows, and raises recall at IoU 0.5 by 0.0193 to 0.158. This repair is a localization coverage diagnostic rather than pairwise caption accuracy, so the lane is evidence for limited practical recovery, not for a solved detector pipeline. The non-VAW families remain a current-schema no-go for practical localization because their task rows do not provide source regions for target-object crop construction.

Oracle-region upper bound. The VAW oracle condition supplies gold object regions and then evaluates object-local scoring on those regions. This is not a practical detector result. It is an upper-bound localization probe: if the oracle lane improves over no-region object-local readout, then the measured gap is consistent with region quality or localization being a binding bottleneck. The paper therefore treats it as evidence for localization sensitivity, not as an end-to-end method.

Qwen/VLM lane and reversal audit. The Qwen/VLM lane chooses between the two alternatives using a black-box VLM answer. The full configured-order run has every gold answer in the first candidate position, so we report it with an explicit candidate-order caveat. A bounded reversed-order audit flips candidates on 320 rows, 80 per family, and checks whether the model continues to choose the content-correct answer. That audit is useful evidence against a simple first-position-only explanation, but it is not a full position-balanced estimate.

Interpretation binding. The method is coupled to an interpretation summary rather than only to a leaderboard.

Each major interpretation is assigned one of four statuses: supported, limited, weak, or rejected. Supported interpretations require a completed local result and a role in the argument. Limited and weak interpretations require scope language. Rejected interpretations are reported when they are scientifically informative but cannot be promoted as contributions. This is why the VAW oracle result can be stated strongly while detected-region recovery is reported only with its coverage boundary.

Why pairwise accuracy. Pairwise accuracy is intentionally simple. The benchmarks already define minimally contrasting alternatives, and the paper’s question is whether a scoring lane ranks the correct alternative above the binding alternative. More complex aggregate scores would make the audit harder to interpret because they could mix calibration, answer style, and confidence scale. Pairwise accuracy also supports exact paired tests whenever two lanes score the same items.

Why oracle regions are isolated. The oracle lane is separated from the no-region object-local lane because they answer different questions. The no-region lane asks whether a train-free local readout helps without privileged region information. The oracle lane asks whether correct localization would be useful if available. Treating them as the same method would overstate the practical evidence. Keeping them separate lets the paper identify localization as a plausible bottleneck while refusing to claim that the bottleneck has been solved.

Interpretation constraints. The method intentionally treats coverage boundaries as hard boundaries. If detected regions cover only a minority of region rows, the oracle result cannot be used to imply a solved practical detector-backed system. If object-local no-region scoring ties global scoring on a family, the paper cannot describe that family as improved by local readout. If a VLM audit covers only a slice, the paper cannot call the full VLM estimate position-balanced. These constraints are simple, but they prevent three common interpretation errors: extrapolating from an oracle to practice, extrapolating from one family to all families, and extrapolating from a slice audit to a full audit.

Why train-free evaluation. Train-free scoring makes the causal interpretation narrower but cleaner. Training interventions can improve a benchmark for many reasons: better visual features, better text features, larger data, or benchmark-specific hard negatives. Here, all completed lanes hold weights fixed and ask what changes when the scoring view changes. The result is not a claim about the best possible model; it is a claim about which evidence is sufficient for a frozen scorer under the measured conditions.

Experimental Setup and Results

The evaluation reports 7,420 scored rows and 22 aggregate condition-family rows. Table 1 summarizes the benchmark families and the way they are used in this study. NLVR2 is included as a replacement/robustness family: it is useful for checking whether a result is limited to direct attribute-swap datasets, but it is not pooled as direct attribute-object evidence.

Table 1: Benchmark setup spans four families and keeps NLVR2 as robustness evidence rather than direct attribute-swap evidence.

Family	Role in study	Paired rows
NLVR2	replacement/robustness	354
SugarCrepe	compositional text alternative	257
Sugar++	lexical/semantic alteration	257
VAW	attribute-object localization	440

The reader-facing system specification is fixed across the result tables. Every benchmark item is a two-candidate ranking problem scored by pairwise accuracy. OpenCLIP lanes use ViT-B/32-style LAION2B S34B B79K image-text scoring; the VAW detected-region lane adds a COCO-pretrained Faster R-CNN ResNet-50-FPN crop selector before the same OpenCLIP scorer and uses full-image fallbacks when the detector abstains; the VAW oracle lane replaces predicted crops with gold object regions; and the Qwen/VLM lane uses temperature-0 eight-token answer generation with one call per item. Paired deltas are computed only for shared task identifiers within the same family.

All aggregate results are computed at the family level. Accuracy intervals are Wilson-style confidence intervals in the reported result table. Paired comparisons use shared task identifiers and report paired bootstrap confidence intervals plus exact McNemar tests. These tests are used only where two methods share the same task items. The main comparisons are global OpenCLIP versus no-region object-local OpenCLIP, VAW detected regions versus global and no-region object-local OpenCLIP, VAW oracle versus VAW no-region and detected object-local scoring, and VAW oracle versus the strongest non-oracle lanes.

Condition roles. The blind text-only condition checks whether the text alternatives alone contain enough artifacts to choose the correct answer. The global-bag condition checks a deliberately weak bag-style control. Global OpenCLIP is the frozen contrastive full-image baseline. Object-local no-region OpenCLIP is the practical train-free local readout that does not use detected regions. Object-local detected-region OpenCLIP is the VAW-only practical localization lane, interpreted with detector coverage and fallback counts. Object-local oracle OpenCLIP is the VAW-only upper-bound probe. Qwen/VLM is a stronger black-box answerer whose configured-order result requires the reversal caveat. These roles prevent the table from being read as a single leaderboard in which all lanes have the same evidential meaning.

Family roles. SugarCrepe and SUGARCREPE++ are the most direct compositional alternatives in the matrix. VAW is the family where object attributes and object localization are most tightly connected to the central claim. NLVR2 is intentionally treated as replacement and robustness evidence: it tests whether the pattern appears outside direct attribute-swap data, but it does not become pooled attribute-object evidence. This distinction matters because over-pooling would make the broad object-local claim look more general than

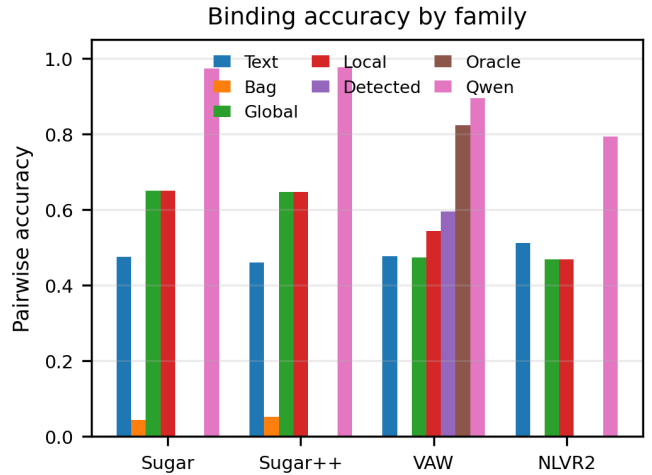


Figure 2: Family-level binding accuracy from the reported result table. Object-local no-region scoring only separates from global OpenCLIP on VAW; detected-region VAW improves over the no-region local lane but remains below the oracle-region upper bound; Qwen/VLM remains high under the stated slice-audit caveat.

the actual paired comparisons support.

Audit policy. A result is promoted only if its caveats are visible in the same section as the number. The Qwen/VLM rows therefore include the candidate-order caveat in the table. The oracle rows include the oracle-region scope. The detected-region VAW row includes the detector-coverage caveat. The no-region object-local rows are not described as a general improvement because the paired deltas do not support that conclusion. This policy prevents a stronger interpretation than the evidence permits.

Table 2 gives the main setup and result matrix. The columns name the evaluated backend, evidential role, metric and budget, and the family-level pairwise accuracy for each benchmark slice. The Qwen/VLM lane is the highest configured-order lane in all four families, but it carries the candidate-order caveat. The no-region object-local lane exactly ties global OpenCLIP on NLVR2, SugarCrepe, and SUGARCREPE++ and differs on VAW. The VAW detected-region and oracle-region lanes are reported only for VAW because this evaluation has detector and oracle region evidence only for that family.

Figure 2 plots the same family-level matrix. Its most important pattern is the separation between global/OpenCLIP-style scoring and the configured-order Qwen/VLM lane, with the oracle VAW point sitting between detected-region object-local OpenCLIP and Qwen/VLM. The plot also makes the rejected claim visible: if object-local readout without regions were a broad fix, its bars would rise above global OpenCLIP across families; instead they coincide on three families.

The reversed-order audit provides a bounded check on the configured-order VLM result. It completes 320 rows, with 282 correct answers and 0.88125 reversed accuracy. The answer distribution changes with the reversal: the model re-

Table 2: Main results: Qwen/VLM is highest in all four families but slice-audited; detected-region VAW improves over no-region and global OpenCLIP but remains coverage-limited; all cells report pairwise accuracy.

Condition	Backend	Role			Metric / budget	NLVR2 n=354	Sugar n=257	Sugar++ n=257	VAW n=440
Blind text-only	OpenCLIP	text	artifact	control	acc.; no image	0.511	0.475	0.459	0.477
Global bag	deterministic	bag	lexical	control	acc.; no model call	0.000	0.043	0.051	0.000
Global OpenCLIP	OpenCLIP B/32	ViT	full-image	baseline	acc.; pair score	0.469	0.650	0.646	0.473
Object-local, no regions	OpenCLIP B/32	ViT	local readout		acc.; no detector	0.469	0.650	0.646	0.543
Object-local, detected regions	OpenCLIP B/32	ViT	practical localization		acc.; detector + fallback	–	–	–	0.595
Object-local, oracle regions	OpenCLIP B/32	ViT	upper bound		acc.; gold VAW boxes	–	–	–	0.823
Qwen/VLM configured	Qwen2.5-VL-3B	VLM	reference		acc.; $T = 0, 8$ tok., 1 call	0.794	0.973	0.977	0.895

turns B on 282 rows and A on 38 rows. Family results are 55/80 on NLVR2 (0.6875), 74/80 on SugarCrep (0.925), 77/80 on SUGARCREPE++ (0.9625), and 76/80 on VAW (0.95). This is inconsistent with a trivial always-answer-A explanation on the audited slice. It is still not a complete balanced rerun, so the paper does not claim a fully position-balanced Qwen/VLM estimate.

The family breakdown also explains why the paper is not framed as a detector-free object-local method. On SugarCrep and SUGARCREPE++, the global and no-region local OpenCLIP lanes are identical at the aggregate level. On NLVR2 they are also identical. Only VAW separates, and that separation is much smaller than the oracle-region separation. The detected-region VAW lane improves over both no-region local and global OpenCLIP, but most region rows fall back to full-image scoring after detector abstention. A method paper would need to show that the local readout generalizes beyond VAW or that detected regions recover more of the oracle gain under stronger coverage. The current evidence does neither, so the claim remains limited rather than a broad method result.

The Qwen/VLM lane gives a different lesson. Its high configured-order accuracy shows that a larger VLM-style answerer can often select the correct alternative in this matrix, and the reversed slice shows that the audited subset does not collapse when the answer position changes. But the full run still has a position-design flaw: every gold answer is in the first candidate position. The paper therefore treats Qwen/VLM as a strong reference lane with a bounded audit, not as a clean final estimate. That stance leaves room for a later full randomized-order rerun without retroactively invalidating the current evidence.

The comparison with blind and bag controls is also informative. Blind text-only accuracy sits near chance across the direct families, which reduces concern that the local benchmark matrix is solved by text artifacts alone. The global-bag control is intentionally weak and sometimes degener-

ates when both candidates contain the same bag of salient words. That degeneration is a feature of the control: it exposes where bag-style evidence is insufficient. The useful lesson is not that the bag control is competitive, but that the stronger lanes must be interpreted against the possibility of lexical and ordering shortcuts.

Analysis, Ablation, and Failure Cases

The paired comparisons in Table 3 sharpen the interpretation. No-region object-local scoring has zero paired delta against global OpenCLIP on NLVR2, SugarCrep, and SUGARCREPE++. On VAW it improves by 0.0705, with a bootstrap interval from 0.0068 to 0.1318 and exact McNemar $p = 0.0377$. Detected regions improve VAW accuracy by 0.1227 over global OpenCLIP and by 0.0523 over no-region object-local scoring, but the oracle lane remains 0.2273 accuracy higher than the detected-region lane. The stronger and cleaner localization result is still the VAW oracle comparison: oracle regions improve over no-region object-local scoring by 0.2795 accuracy with a bootstrap interval from 0.2295 to 0.3318.

Figure 3 visualizes the VAW paired deltas. The oracle comparison is large enough to support a localization-sensitive upper-bound claim. The detected-region comparisons are positive but smaller: practical detected regions recover part of the no-region gap while leaving a large oracle-vs-detected gap. The negative comparison against Qwen/VLM is also informative: the oracle OpenCLIP lane does not beat the configured-order Qwen/VLM lane. This reinforces the narrow interpretation. Oracle regions help the frozen object-local scorer, but they are not evidence that object-local OpenCLIP dominates modern VLM answering.

The ablation structure is therefore asymmetric. The no-region object-local lane is an ablation of global scoring that changes readout locality without adding gold regions. The detected-region lane adds an automatic localization step but retains full-image fallbacks when the detector abstains. The

Table 3: Paired tests support the VAW oracle localization upper bound, show limited detected-region recovery, and reject a broad no-region object-local improvement.

Comparison	Family	Shared rows	Delta	95% CI	McNemar p
Global vs. object-local no regions	NLVR2	354	0.0000	[0.0000, 0.0000]	1.0000
Global vs. object-local no regions	SugarCrepe	257	0.0000	[0.0000, 0.0000]	1.0000
Global vs. object-local no regions	SUGARCREPE++	257	0.0000	[0.0000, 0.0000]	1.0000
Global vs. object-local no regions	VAW	440	0.0705	[0.0068, 0.1318]	0.0377
VAW global vs. detected regions	VAW	440	0.1227	[0.0591, 0.1841]	0.0003
VAW no-region vs. detected regions	VAW	440	0.0523	[0.0159, 0.0909]	0.0067
VAW no-region vs. oracle regions	VAW	440	0.2795	[0.2295, 0.3318]	1.87×10^{-22}
VAW detected vs. oracle regions	VAW	440	0.2273	[0.1773, 0.2773]	3.41×10^{-17}
Qwen/VLM vs. VAW oracle regions	VAW	440	-0.0727	[-0.1159, -0.0295]	0.0016

oracle-region lane is an upper-bound ablation that changes the quality of localization evidence. The paired tests show that changing locality alone is not enough to obtain broad gains, practical detected regions recover only part of the VAW gap, and supplying correct regions on VAW is enough to obtain a large gain over no-region object-local scoring. This is the core mechanism inference the paper can support.

The analysis also identifies where more experiments would be valuable. The detected-region run answers whether a first practical detector recovers any VAW oracle gain, but its coverage is low even after the repaired mapping: 321 of 880 region rows use detector-matched crops, 559 remain fallback or missing rows, and recall@0.5 is 0.15795454545454546. The improvement over the previous 208/880 detected baseline is a localization-coverage delta of 0.019318181818181818, not new pairwise caption accuracy. The next detector question is therefore not whether any detected-region evidence exists, but how much of the oracle gain survives when detector recall and category coverage improve. If a full randomized candidate-order VLM run were available, the next question would be whether the configured-order Qwen/VLM accuracies remain high across all 1,308 VLM rows.

The VAW result is the most important positive finding because it has the right paired structure for the localization question. The no-region object-local lane and the detected-region and oracle-region lanes share the same 440 VAW tasks, so the paired deltas are not cross-dataset comparisons. The oracle bootstrap interval remains far from zero, and the exact McNemar test identifies many more items corrected by the oracle than lost by it. This supports the claim that correct object localization can matter substantially for attribute-object binding under frozen scoring.

The broad object-local result is the most important negative finding because it blocks an overgeneralized method story. If local readout alone were enough, the SugarCrepe, SUGARCREPE++, and NLVR2 rows would move. They do not. This negative finding does not make the VAW oracle result less real; it says that the localization advantage requires actual region evidence in the measured setup. The detected-region lane confirms that imperfect automatic regions can recover a smaller part of this effect on VAW, but it also shows that detector coverage is now the practical bottleneck. That

distinction is the paper’s main analytical contribution.

This distinction also changes how the reader should interpret the benchmark families. SugarCrepe and SUGARCREPE++ are difficult because their alternatives are designed to reduce shallow caption artifacts. A local readout that exactly tracks global OpenCLIP there is not a failed implementation detail; it is evidence that this particular local readout did not add useful binding information beyond whole-image contrastive similarity on those families. NLVR2 plays a different role. Its replacement-style examples test whether the same lanes behave sensibly outside direct attribute swaps, but they should not be pooled into an attribute-binding headline. The VAW result is narrower and more mechanistic because it has object attributes and oracle regions aligned to the question the paper asks.

The same logic applies to the Qwen/VLM result. A high answerer score is useful for contextualizing the difficulty of the matrix, but the paper does not use it to erase the OpenCLIP analysis. The Qwen/VLM lane answers a different question: whether a stronger black-box VLM can often choose the right option in the configured benchmark order and in a bounded reversed slice. The OpenCLIP lanes answer whether train-free global versus local contrastive evidence changes binding accuracy. Keeping those questions separate avoids a misleading conclusion in either direction. The Qwen/VLM lane does not invalidate the oracle-region upper bound, and the oracle-region upper bound does not imply that OpenCLIP object-local scoring outperforms Qwen/VLM.

The final scope decision is to keep the detected-region bottleneck in the main analysis rather than burying it after the bibliography. Detected regions are the bridge between an oracle localization insight and a practical method, and the measured bridge is narrow in this run. The repaired localization stratum covers 321 region rows, while 559 region rows remain fallback or missing because the object name is unmapped after expansion or the mapped category has no detection. This makes the detected result scientifically meaningful but limited: it is a measured partial recovery, not an end-to-end localization solution.

Table 4 summarizes how the main measurements constrain the interpretation. The VAW oracle result is the clearest localization signal, the detected-region lane shows par-

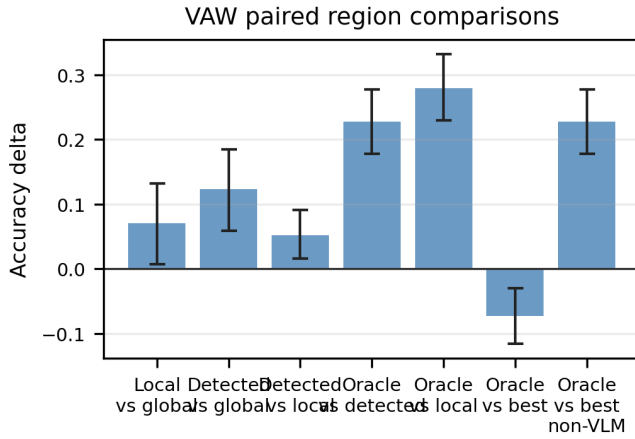


Figure 3: VAW paired deltas separate the supported oracle-region upper bound, the smaller detected-region recovery, and the unsupported broad recovery story.

tial but coverage-limited VAW recovery, and the no-region local readout does not provide a broad cross-family fix. The Qwen/VLM lane supplies a strong reference point under an explicit order audit.

The measured detected-region lane is the main practical boundary. The evaluation does not fabricate detector boxes: it uses detector-matched crops when available and full-image fallbacks when the detector abstains. As a result, the VAW oracle delta identifies what correct regions can make possible for a frozen scorer, while the detected-region row shows what the current detector-covered system recovers automatically. A detector follow-up should be judged by how much of the remaining 0.2273 oracle-vs-detected gap closes when regions are predicted with better target coverage.

The controls define a different boundary. The blind text-only lane checks whether the alternatives carry enough linguistic bias to solve the item without visual evidence. The global-bag lane checks whether unordered lexical overlap collapses the decision. These controls are not intended to be competitive vision-language methods; they make the stronger lanes accountable to shortcut explanations. A future method that improves over OpenCLIP but fails against these controls would require a different interpretation.

The Qwen/VLM audit defines the remaining boundary. The reversed slice is inconsistent with a simple first-position rule on the audited rows, but it is still a slice. A full candidate-order experiment could change family-level estimates and answer-position distributions. The present comparison therefore uses Qwen/VLM to contextualize benchmark difficulty while keeping the OpenCLIP localization analysis separate.

Conclusion

This study reports a diagnostic no-go that the measured evidence can support. Train-free binding evaluation shows a strong VAW oracle-region localization upper bound, rejects broad no-region object-local improvement, and re-

Table 4: Evidence-to-interpretation summary: the VAW oracle result is a localization upper bound, while detected-region recovery is measured but coverage-limited.

Finding	Evidence	Scope boundary
VAW localization	Oracle regions reach 0.823 vs. 0.543 no-region local accuracy	Upper bound, not detector recovery
Detected-region VAW	Detected regions reach 0.595; repaired localization coverage reaches 321/880 matched rows	559 fallback/missing rows; localization coverage, not pairwise caption accuracy
No-region local readout	Ties global OpenCLIP on NLVR2, SugarCrepe, and SUGARCREPE++	Not a broad local fix
Qwen/VLM reference	0.794–0.977 configured accuracy; 0.88125 accuracy on 320 reversed rows	Not fully randomized by position
Shortcut controls	Blind and bag lanes expose text and lexical shortcuts	Controls, not competitive methods

ports Qwen/VLM performance as slice-audited rather than fully position-balanced. The practical route is not solved: detected regions provide measured but coverage-limited VAW recovery, remain far below the oracle upper bound, and cannot be broadened to the selected non-VAW families under the current source-region schema. The repaired coverage audit matches 321/880 region rows, leaves 559 fallback or missing rows, and recall@0.5 remains 0.15795454545454546. Future work should add train-free target-object proposal/source-region evidence for non-VAW families and run a full position-balanced VLM audit before making a stronger practical recovery claim.

Limitations

The main limitation is the gap between oracle localization and practical localization. The VAW oracle-region lane uses gold object regions and therefore tests whether correct localization would help a frozen object-local scorer. It does not show that a detector can recover the same gain. This distinction is central to the result: Table 3 supports a 0.2795 VAW oracle delta over the no-region object-local lane, while the detected-region lane recovers 0.0523 over no-region local scoring but remains 0.2273 below the oracle lane. The detector coverage explains the gap: the repaired localization-coverage audit matches 321 of 880 region rows, leaves 559 fallback or missing rows, and measures recall@0.5 at 0.15795454545454546, only 0.019318181818181818 above the previous detected baseline.

The Qwen/VLM reference lane is also bounded. The configured-order run covers all four reported families, but every gold answer appears in the first candidate position. The 320-row reversed-order audit is useful because the

model remains accurate on the audited slice and mostly moves to the new correct position, but the audit is not a full randomized-order evaluation. The paper therefore uses Qwen/VLM as a strong reference lane rather than as a fully position-balanced estimate.

Benchmark scope is a third boundary. SugarCrepe and SUGARCREPE++ provide direct compositional caption alternatives, VAW provides object-attribute localization structure, and NLVR2 is included as replacement/robustness evidence. These families are enough to reject a broad no-region object-local claim in the measured setup, but they do not establish a universal account of all attribute-object binding failures. The reported negative result should be read as evidence about these frozen scoring lanes and task slices, not as a claim that localization is irrelevant in general.

The negative-family pattern is especially important for interpreting the object-local lane. On SugarCrepe, SUGARCREPE++, and NLVR2, the no-region local readout exactly matches the global OpenCLIP aggregate in Table 2, and the paired deltas in Table 3 are zero for the three shared families. That outcome does not prove that local evidence is useless. It shows that this train-free no-region readout did not add binding-relevant information beyond the full-image contrastive score on those slices. The VAW-only improvement is therefore not a license to generalize; it is a pointer to the interaction between object attributes, region evidence, and the scorer.

The VAW oracle result also has an asymmetric interpretation. It is stronger than the no-region local readout because it supplies the object regions the binding question needs, but it is weaker than a practical method because those regions are privileged. The detected-region lane provides that practical bridge only partially. It reports the same paired VAW tasks under predicted regions and fallbacks, but the low matched-coverage rate means the present evidence identifies a localization bottleneck without showing that the bottleneck has been solved.

Finally, the VLM audit should be interpreted as a guardrail rather than a complete robustness study. The reversed slice rules out the simplest always-answer-A explanation on the audited rows, but a full randomized-order rerun could still change family-level estimates, especially on NLVR2 where the reversed slice is weaker than the direct compositional families. The next VLM validation should randomize candidate order for all 1,308 VLM-scored rows and report both family accuracy and answer-position distribution. Until then, the Qwen/VLM lane is useful context for benchmark difficulty, not a replacement for the paired OpenCLIP localization analysis.

Ethical Considerations

This work is diagnostic and does not train or deploy a model, but its interpretation can still affect downstream claims about vision-language systems. The main ethical risk is overclaiming: an oracle-region improvement could be mistaken for solved detector-backed recovery, a coverage-limited detected-region gain could be mistaken for deployment-ready grounding, or a high VLM score could

be mistaken for a fully audited robustness result. The paper mitigates that risk by keeping detector coverage, the candidate-order caveat, and the benchmark-family scope visible in the main text. The evaluation also uses public vision-language benchmarks whose annotations and images may reflect dataset collection biases; the reported results should therefore be treated as benchmark diagnostics rather than statements about real-world visual understanding across populations or contexts.

The evaluation design also has communication risks. Pairwise caption benchmarks can reward systems that exploit language priors, dataset regularities, or answer format conventions, and the candidate-order audit only checks a bounded slice of the VLM reference lane. We therefore avoid using the Qwen/VLM accuracy as a deployment-quality ranking and report it only as context for how difficult the same pairs are for a strong black-box answerer. Similarly, the oracle VAW lane uses privileged regions and should not be read as evidence that an automatic region proposal system would recover the same gain; the measured detected-region lane shows only partial recovery under limited detector coverage.

These boundaries matter because diagnostic papers can influence what follow-up work chooses to measure. Reporting the failed no-region local lane, the limited detected-region lane, and the oracle lane together is intended to make the remaining detector bottleneck visible rather than to advertise a solved localization method. Any downstream use of this benchmark package should rerun the paired protocol on the target data and separate oracle evidence, detected-region evidence, and black-box answerer evidence before making claims about practical visual grounding.

Reproducibility Statement

The reported study is train-free and inference-only. The evaluated lanes are defined by the benchmark family, answer format, model backend, region source, and paired scoring rule rather than by learned parameters introduced in this paper. The global and object-local contrastive lanes use the same OpenCLIP ViT-B/32-style scoring backend, with the local variants changing whether the image evidence is the whole image, no-region object text, a VAW detected crop or full-image fallback, or the VAW oracle region. The Qwen/VLM lane uses the same pairwise answer format with temperature zero and an eight-token answer budget. These settings make the negative, limited, and positive comparisons reproducible as evaluation conditions rather than as training recipes.

The main result table reports every completed family-by-method cell used for the claims, and the paired table reports the shared-task comparisons used to interpret localization. The family row counts distinguish direct compositional-caption tests from the VAW oracle-region slice and from NLVR2 replacement/robustness evidence. They also distinguish detected-region VAW from oracle-region VAW because detector coverage changes the interpretation. The reversal audit for Qwen/VLM is reported as a bounded diagnostic slice, not as a complete

position-balanced rerun. A replication should therefore preserve three separations: global versus object-local scoring, no-region/detected/ oracle-region evidence on VAW, and configured-order versus reversed-order VLM answering.

The aggregation rule is also part of the result. Accuracy is computed at the pair level within each benchmark family, and paired deltas are computed only where the same task identifiers appear in both compared lanes. This prevents the VAW oracle result from being pooled directly with non-oracle families and prevents the Qwen/VLM reference lane from being treated as a paired localization intervention. Reproducing the headline conclusions therefore requires preserving family-level reporting, the shared-task paired tests, and the distinction between diagnostic controls and evaluated vision-language backends.

For detector-backed follow-up work, the same reporting discipline is necessary because a detector can fail before the scorer sees the intended object. A practical localization run should therefore separate proposal recall, wrong-box selection, scorer errors given a correct box, fallback rows, and final pair accuracy. Reporting only the final pair accuracy would hide whether the bottleneck is visual proposal quality or object-attribute scoring. For randomized-order VLM follow-up work, the answer-position distribution should be reported together with family accuracy so that high accuracy cannot be explained by a residual format prior.

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Appendix: Protocol Notes

The appendix records three protocol details not needed for the main claim. First, the Qwen/VLM reversal audit flips the candidate order on 80 rows per family and keeps the same pairwise answer format. Second, the oracle-region condition is available only for VAW in this evidence package because the other reported families do not provide the same verified object-region interface. Third, the detected-region condition is also VAW-only here and separates detector-matched crops from full-image abstention fallbacks.